Lecture 11: Lagrangians (cont'd) and Expenditure Minimization

1 Where are we?

- Last time, we stated the consumer problem,

  established a few properties of Marshallian demand and the indirect utility function,

  introduced Lagrangians as a way to solve the consumer problem,

  and did most of the work to establish the relationship between solutions to the consumer problem and saddle points of the Lagrangian

- Today, we wrap up that result,

  see the Kuhn-Tucker conditions,

  and see examples of how to use this stuff

- First... any questions?

2 Recap

- So, we have the Consumer Problem,

$$\max_{x \in \mathbb{R}^k} u(x) \qquad \text{subject to} \qquad p \cdot x \leq w \quad \text{and} \quad x \geq 0$$

- We defined the Lagrangian,

$$\mathcal{L} \quad = \quad u(x) + \lambda(w - p \cdot x) + \mu \cdot x$$

  defined for $x \in \mathbb{R}^k$ and $(\lambda, \mu) \in \mathbb{R}^{k+1}_+$

- A saddle point of the Lagrangian is a point (x^*, λ^*, μ^*) such that

$$\mathcal{L}(x, \lambda^*, \mu^*) \quad \leq \quad \mathcal{L}(x^*, \lambda^*, \mu^*) \quad \leq \quad \mathcal{L}(x^*, \lambda, \mu)$$

  for any $x \in \mathbb{R}^k$ and any $(\lambda, \mu) \in \mathbb{R}^{k+1}_+$

- We proved that if (x^*, λ^*, μ^*) is a saddle point of the Lagrangian,
  then x^* is a solution to the consumer problem

  - Key steps of the proof:
    we showed that if (x^*, λ^*, μ^*) is a saddle point, then x^* must be feasible;
    and that complementary slackness must hold, $\lambda^*(w - p \cdot x^*) = \mu^* \cdot x^* = 0$;
    that implied that $\mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*)$, since the other terms in the Lagrangian are 0;
    and this implied that for any other x' that's feasible,

$$u(x^*) \quad = \quad \mathcal{L}(x^*, \lambda^*, \mu^*) \quad \geq \quad \mathcal{L}(x', \lambda^*, \mu^*) \quad \geq \quad u(x')$$

    so $u(x^*) \geq u(x')$ for any x' that's feasible,
    so x^* solves the consumer problem

- We claimed a version of the converse:

  that if u is differentiable and concave and x^* solves the consumer problem,

  we can find λ^* and μ^* such that (x^*, λ^*, μ^*) is a saddle point of $\mathcal{L}$

- To do this, we picked a good j for which $x_j^* > 0$, and defined

$$\lambda^* \quad = \quad \frac{1}{p_j} \frac{\partial u}{\partial x_j}(x^*)$$

  and then defined

$$\mu_i^* \quad = \quad \begin{cases} 0 & \text{if} \quad x_i^* > 0 \\ \\ \lambda^* p_i - \dfrac{\partial u}{\partial x_i}(x^*) & \text{if} \quad x_i^* = 0 \end{cases}$$

- We showed that λ^* defined in this way is weakly positive;

  that it was uniquely defined, i.e., would have been the same for any j with $x_j^* > 0$;

  and that μ^* defined in this way is weakly positive

- What's left is to show that (x^*, λ^*, μ^*) is a saddle point

- We showed the first part – that for any x,

$$\mathcal{L}(x^*, \lambda^*, \mu^*) \quad \geq \quad \mathcal{L}(x, \lambda^*, \mu^*)$$

  by showing that λ^* and μ^* were defined to make $\frac{\partial \mathcal{L}}{\partial x_i} = 0$ at (x^*, λ^*, μ^*),

  and since $\mathcal{L}$ is differentiable and concave, the FOC implies it's maximized at x^*

- What's left, then, is to show that for any $\lambda, \mu \geq 0$,

$$\mathcal{L}(x^*, \lambda^*, \mu^*) \quad \leq \quad \mathcal{L}(x^*, \lambda, \mu)$$

  which will establish we've found a saddle point

- First, note that the way we defined λ^* and μ^*, they must satisfy complementary slackness
  - If $w - p \cdot x^* = 0$, then $\lambda^*(w - p \cdot x^*) = 0$, of course
  - If $w - p \cdot x^* > 0$, it means the consumer optimally spend less than their full budget, which means they couldn't have increased u by consuming a little more of good j, but also couldn't have increased u by consuming less of good j, meaning $\frac{\partial u}{\partial x_j}(x^*) = 0$, so then $\lambda^* = 0$ by construction
  - And we defined μ_i^* to be 0 whenever $x_i^* \neq 0$, so $\mu_i^* x_i^* = 0$ also by construction

- Now, at $x = x^*$, $\mathcal{L}$ is linear in λ, with slope $w - p \cdot x^*$
  meaning if $w - p \cdot x^* = 0$, any λ gives a minimum, so this holds;
  and if $w - p \cdot x^* > 0$, we've set $\lambda^* = 0$, so $\lambda = \lambda^*$ minimizes $\mathcal{L}$

- Similarly, at $x = x^*$, $\mathcal{L}$ is linear in μ_i, with slope x_i^*;
  if $x_i^* = 0$, then any value of μ_i minimizes $\mathcal{L}$,
  and if $x_i^* > 0$, $\mathcal{L}$ is increasing in μ_i and $\mu_i^* = 0$,
  so μ_i^* minimizes $\mathcal{L}$ either way

- So indeed, (λ^*, μ^*) solves $\min \mathcal{L}(x^*, \lambda, \mu)$

- So that's the result –
  if u is differentiable and concave and $(p, w) \gg 0$,
  then for any solution to the consumer problem, there's a saddle point of $\mathcal{L}$

3 The Kuhn-Tucker Conditions

- So, we know that the solutions to the Consumer Problem
  pretty much correspond to the saddle points of the Lagrangian
- Next, let's think about what conditions (x^*, λ^*, μ^*) must satisfy to be a saddle point,
  and therefore for x^* to solve the consumer problem
- First, we need
$$x^* \quad \in \quad \arg\max_{x \in \mathbb{R}^k} \mathcal{L}(x, \lambda^*, \mu^*)$$
- Note that since we're thinking of $\mathcal{L}$ as defined for all x in $\mathbb{R}^k$, not just $\mathbb{R}^k_+$,
  x^* must be an interior point;
  if u is differentiable, and therefore $\mathcal{L}$ is differentiable w.r.t. x, we'll need
$$\frac{\partial \mathcal{L}}{\partial x_i}(x^*, \lambda^*, \mu^*) \quad = \quad 0$$
  or
$$\frac{\partial u}{\partial x_i}(x^*) - \lambda^* p_i + \mu_i^* \quad = \quad 0$$
- (If this was either positive or negative, x^* couldn't be maximizing $\mathcal{L}$)
- And second, we need
$$(\lambda^*, \mu^*) \quad \in \quad \arg\min_{\lambda, \mu \geq 0} \mathcal{L}(x^*, \lambda, \mu) \quad = \quad \arg\min_{\lambda, \mu \geq 0} \{u(x^*) + \lambda(w - p \cdot x^*) + \mu \cdot x^*\}$$
- This means we need $w - p \cdot x^* \geq 0$ – otherwise, we could always reduce $\mathcal{L}$ by increasing λ –
  and we need $\lambda^* = 0$ if $w - p \cdot x^* > 0$, otherwise we could reduce $\mathcal{L}$ by reducing λ
- And similarly, we need $x^* \geq 0$ – if $x_i^* < 0$, we could reduce $\mathcal{L}$ by increasing μ_i –
  and we need $\mu_i^* = 0$ if $x_i^* > 0$, or else we could reduce $\mathcal{L}$ by reducing μ_i
- So basically, for (λ^*, μ^*) to minimize $\mathcal{L}$,
  we need for x^* to be feasible, and we need complementary slackness to hold

- And of course, we defined $\mathcal{L}$ for non-negative values of the Lagrange multipliers, so we can think of $\lambda \geq 0$ and $\mu \geq 0$ as additional constraints
- That gives us a bunch of necessary conditions for (x^*, λ^*, μ^*) to be a saddle point:
  1. $\frac{\partial u}{\partial x_i}(x^*) - \lambda^* p_i + \mu_i^* = 0$ for each i (FOC w.r.t. x)
  2. $p \cdot x^* \leq w$, $x_i^* \geq 0$ for each i (original constraints on x)
  3. $\lambda^* \geq 0$ and $\mu_i^* \geq 0$ for each i (non-negative Lagrange multipliers)
  4. $\lambda^*(w - p \cdot x^*) = 0$ and $\mu_i^* x_i^* = 0$ for each i (complementary slackness)
- These are collectively known as the **Kuhn-Tucker Conditions**
- As expected, these are necessary conditions for a solution to the consumer problem, and also sufficient under some additional assumptions:

  Theorem 1. *Assume u is differentiable and $(p, w) \gg 0$. If $x^* > 0$ solves the Consumer Problem, then there exist $\lambda^* \geq 0$ and $\mu^* \geq 0$ such that (x^*, λ^*, μ^*) satisfy the Kuhn-Tucker conditions.*[1]

  Theorem 2. *Suppose u is differentiable and concave.*[2] *Then if (x^*, λ^*, μ^*) satisfies the Kuhn-Tucker conditions, x^* solves the Consumer Problem.*

- This gradient condition in Theorem 2 holds automatically when u is concave, so when u is differentiable and concave, we have

$$\begin{gathered}(x^*, \lambda^*, \mu^*) \text{ a saddlepoint of } \mathcal{L} \text{ for some } \lambda^*, \mu^* \\ \updownarrow \\ x^* \text{ solves the consumer problem} \\ \updownarrow \\ (x^*, \lambda^*, \mu^*) \text{ satisfies the Kuhn-Tucker conditions for some } \lambda^*, \mu^*\end{gathered}$$

- (If u is differentiable but not concave, we still have $\mathcal{L} \to CP \to KT$ – the downward arrows still hold – and $\mathcal{L} \to CP$ doesn't even require differentiability, but the upward arrows require concavity or something like it)

[1] For a more general constrained optimization problem, there is one additional restriction which would have to hold at x^*, in order to guarantee the Kuhn-Tucker conditions can be satisfied. This is called *constraint qualification*, and it basically says that whichever of the constraints of the problem bind at x^*, their gradients need to be linearly independent, i.e., there can't be any locally "redundant" constraints. Given the constraints in our problem (budget and non-negativity), this condition automatically holds whenever $p \gg 0$ and $w > 0$, so I don't state it in the theorem; this is just something you need to know if you want to extend these techniques to other problems. See MWG page 956-964 for the more general treatment of constrained optimization.

[2] We can weaken concavity to (i) quasi-concavity and (ii) for any $x, x' \in X$, $u(x') > u(x)$ implies $\nabla u(x) \cdot (x' - x) > 0$, both of which hold when u is concave.

4 What's this good for?

- One nice thing: even if u is not concave

  (so the KT conditions aren't automatically sufficient),

  the KT conditions will often have a unique solution;

  since the CP must have a solution, and it must satisfy KT,

  if the KT conditions have a unique solution, we know it's the unique solution to the CP!

- Let's use this to solve one example
- Let $X = \mathbb{R}^3_+$, $u(x, y, z) = x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3}$, a generic price vector $(p_1, p_2, p_3) \gg 0$, and $w > 0$
- First, it's helpful to try to eliminate some constraints
- Note that if any consumption quantity is 0, then $u = 0$, so this can't be optimal –

  so $x \gg 0$, and $\mu = 0$, so we can ignore the non-negativity constraints
- (For many utility functions, the marginal utility of each good is infinite when you're consuming none of it –

  the first little bit of each good is extremely valuable –

  and in this case, it's never optimal to consume 0 of anything given positive wealth,

  and you can just solve the problem without the nonnegativity constraints)

- Next, rather than $u = x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3}$, let's maximize $\ln u$, since this is another utility function representing the same preferences, so we'll use

$$\tilde{u}(x, y, z) \quad = \quad \alpha_1 \log x_1 + \alpha_2 \log x_2 + \alpha_3 \log x_3$$

- The Lagrangian is therefore

$$\mathcal{L} \quad = \quad \alpha_1 \log x_1 + \alpha_2 \log x_2 + \alpha_3 \log x_3 + \lambda(w - p \cdot x)$$

  and the Kuhn-Tucker FOC are therefore

$$\lambda p_i \quad = \quad \frac{\partial u}{\partial x_i} \quad = \quad \frac{\alpha_i}{x_i} \qquad \rightarrow \qquad x_i \quad = \quad \frac{\alpha_i}{\lambda p_i}$$

  for each good i

- Since preferences are LNS, Walras' Law holds, so

$$w \quad = \quad \sum_{j=1}^{3} p_j x_j \quad = \quad \sum_{j=1}^{3} \frac{\alpha_j}{\lambda}$$

so $\lambda = \frac{1}{w} \sum_j a_j$, and therefore

$$x_i \quad = \quad \frac{\alpha_i}{\lambda p_i} \quad = \quad \frac{\alpha_i}{\sum_j \alpha_j} \frac{w}{p_1}$$

or

$$p_i x_i \quad = \quad \frac{\alpha_i}{\alpha_1 + \alpha_2 + \alpha_3} w$$

- Since this is the only way to satsify the K-T conditions,[3]

  we know this is the (unique) solution to the consumer problem!

- (also note that we didn't rely on concavity of u –

  in fact, we never even specified that $\alpha_1 + \alpha_2 + \alpha_3 \leq 1$!)

[3]Walras' Law isn't one of the KT conditions, but since $\frac{\partial u}{\partial x_i} > 0$ at any potentially-optimal point x, we know $\lambda^* > 0$, so $p \cdot x^* = w$ is required to satisfy complementary slackness.

5 Back to saddle points

- Now, (x^*, λ^*, μ^*) being a saddle point of $\mathcal{L}$ is really two separate conditions:
  - given λ^* and μ^*, x^* maximizes $\mathcal{L}$
  - and given x^*, the multipliers (λ^*, μ^*) minimize $\mathcal{L}$
- And we've seen that the latter condition is really equivalent to two things:

  feasibility of x^*, and complementary slackness
- So when I learned this stuff, I basically learned it as,

  you solve a constrained optimization problem by solving

$$\max_x \{u(x) + \lambda(w - p \cdot x) + \mu \cdot x\}$$

  and imposing the additional constraints that $p \cdot x \leq w$, $x \geq 0$, and complementary slackness

- I also mentioned before that if (x^*, λ^*, μ^*) is a saddle point of $\mathcal{L}$,

  then x^* solves $\max_x \mathcal{L}(x, \lambda^*, \mu^*)$,

  and (λ^*, μ^*) solves

$$\min_{\lambda,\mu \geq 0} \max_x \mathcal{L}(x, \lambda, \mu)$$

- And since complementary slackness holds at a saddle point,

$$\mathcal{L}(x^*, \lambda^*, \mu^*) \quad = \quad u(x^*) + \lambda^*(w - p \cdot x^*) + \mu^* \cdot x^* \quad = \quad u(x^*) \quad = \quad v(p, w)$$

- So we can write

$$v(p, w) \quad = \quad \min_{\lambda,\mu \geq 0} \max_x \mathcal{L}(x, \lambda, \mu)$$

- What's nice about this is that we've written $v(p, w)$ as the solution to an *unconstrained* optimization problem – which will be nice if we want to, say, apply the Envelope Theorem to it or something...

6 Wait, what?

- To show you what I mean, I'll use the Envelope Theorem to prove a cool result:

  Theorem 3 (Roy's Identity). *Suppose v is differentiable at $(p, w) \gg 0$, and $\partial v/\partial w > 0$. Then $x(p, w)$ is a singleton and*

$$x_i(p, w) \quad = \quad -\frac{\partial v}{\partial p_i}(p, w) \Big/ \frac{\partial v}{\partial w}(p, w)$$

- Starting with

$$v(p, w) \quad = \quad \min_{\lambda,\mu \geq 0} \max_x \{u(x) + \lambda(w - p \cdot x) + \mu \cdot x\}$$

  we'll let

$$\Phi(\lambda, \mu, p, w) \quad = \quad \max_x \{u(x) + \lambda(w - p \cdot x) + \mu \cdot x\}$$

  be the result of the inner maximization problem, so that

$$v(p, w) \quad = \quad \min_{\lambda,\mu \geq 0} \Phi(\lambda, \mu, p, w)$$

- By the envelope theorem,

$$\frac{\partial v}{\partial w} \quad = \quad \left.\frac{\partial \Phi}{\partial w}\right|_{\lambda=\lambda^*,\mu=\mu^*}$$

- And applying the envelope theorem again,

$$\frac{\partial \Phi}{\partial w} \quad = \quad \left.\frac{\partial \mathcal{L}}{\partial w}\right|_{x=x^*}$$

- So now

$$\frac{\partial v}{\partial w} \quad = \quad \frac{\partial}{\partial w}\left(u(x) + \lambda(w - p \cdot x) + \sum_i \mu_i x_i\right)\Bigg|_{\lambda=\lambda^*,\mu=\mu^*,x=x^*} \quad = \quad \lambda|_{\lambda=\lambda^*} \quad = \quad \lambda^*$$

- This is interesting in its own right – if (x^*, λ^*, μ^*) is a saddlepoint, $dv/dw = \lambda^*$ –
  the value of the multiplier on the budget constraint, is the marginal value of wealth!
  (That is, the Lagrange multiplier tells you the marginal benefit of relaxing that constraint!)

- Then we can do the same thing differentiating with respect to price p_i, and find

$$\frac{\partial v}{\partial p_i} = \frac{\partial}{\partial p_i}\left(u(x) + \lambda(w - p \cdot x) + \sum_i \mu_i x_i\right)\Bigg|_{\lambda=\lambda^*,\mu=\mu^*,x=x^*} = -\lambda x_i|_{\lambda=\lambda^*,x=x^*} = -\lambda^* x_i^*$$

- So

$$-\frac{\partial v/\partial p_i}{\partial v/\partial w} = -\frac{-\lambda^* x_i^*}{\lambda^*} = x_i^*$$

  which is Roy's Identity

7 The Expenditure Minimization Problem

- Next, we're going to introduce a different formulation of the consumer problem, look at its properties, and then look at the link between the two
- In producer theory, we considered the general problem of profit maximization, and we also considered the more specialized problem of cost minimization
- Here, we're going to do something very analogous
- We defined

$$v(p, w) = \max u(x) \quad \text{subject to} \quad p \cdot x \leq w$$

  as the value of the consumer problem, and Marshallian demand $x(p, w)$ as the solution
- The Expenditure Minimization Problem – or dual consumer problem – is defined as

$$\min_{x \geq 0} p \cdot x \quad \text{subject to} \quad u(x) \geq u$$

- That is, rather than asking how much utility we can buy for wealth w, we ask how cheaply we can achieve the utility level u
- We call $e(p, u)$ the min – the smallest amount of money that can achieve utility u – and $h(p, u)$ the solution to the problem, which we'll call Hicksian demand, or compensated demand

8 Why is this problem interesting?

- Our original consumer problem seems like a reasonable proxy for how people might actually make decisions
- The expenditure minimization problem, less so.
- So why do we like it?

- Three reasons
- First, Marshallian Demand faces two effects when a price changes –
  substitution and wealth effects
  (If p_1 goes down, you might buy less of another good and more of good 1
  But in addition, if p_1 goes down, this makes you effectively richer –
  which can change your demand in any direction)
  Hicksian demand eliminates the wealth effect –
  so it gives a way to kind of decompose price effects into their separate components

- Second, because Hicksian demand doesn't have wealth effects,
  we can get stronger, testable predictions
  And importantly, we can infer Hicksian demand from Marshallian demand –
  so if we observe data from the regular consumer problem,
  this gives us a way to test the predictions of the model

- Third, we can't really do welfare analysis using indirect utility functions,
  because utility has no natural scale and people aren't comparable
  But the Expenditure Function is in dollars –
  we can measure welfare changes by how much money it would take to compensate people,
  and then aggregate across people in a more valid way

- (It also creates a nice parallel with what we did in producer theory)

9 Does the Expenditure Minimization Problem Have a Solution?

- With Marshallian demand, when $p \gg 0$, we noted that $B(p, w)$ is compact,
  so we knew the consumer problem has a solution
- With expenditure minimization, we're minimizing over $\mathbb{R}^k_+$ –
  or at least the part with u high enough –
  which is unbounded
- So it's not immediately obvious a solution exists
- Turns out, though, it does:
- **Theorem.** If $p \gg 0$, u is continuous, and there exists an x with $u(x) \geq u$, then the expenditure minimization problem has a solution.
  - Proof. Fix $\hat{x}$ with $u(\hat{x}) \geq u$. Define

    $$S \quad = \quad \{x | u(x) \geq u\} \cap \{x | p \cdot x \leq p \cdot \hat{x}\}$$

    the intersection of the set of points better than u and the set of points cheaper than $\hat{x}$
  - The first set is closed, because preferences are continuous;
    the second set is closed and bounded, since it's a budget set;
    so the intersection is closed and bounded, therefore compact
  - So a minimum exists over S;
    and the solution over S must solve the original problem (over all $\mathbb{R}^k_+$),
    because the points we've excluded were all strictly more expensive than those in S.
- So, we let $h(p, u)$ be the set of minimizers – the Hicksian, or Compensated, Demand
  And $e(p, u)$ the min value – the Expenditure Function

10 Properties of Hicksian Demand and the Expenditure Function

- Expenditure minimization is the problem of minimizing a linear function $(p \cdot x)$
  over an arbitrary set $(\{x : u(x) \geq x\})$
- Which means it has the exact same structure as a firm's cost minimization problem;
  and the same structure as the firm's profit maximization problem, just with the sign flipped
- So Hicksian demand, and the expenditure function,
  will have a whole bunch of familiar-looking properties
- **Proposition.** Suppose u is a continuous utility function,
  representing a preference relation $\succsim$ on $\mathbb{R}^k_+$.
  1. $h(p, u)$ is homogenous of degree 0 in p – $h(\lambda p, u) = h(p, u)$ for any $\lambda > 0$
  2. if preferences are convex, $h(p, u)$ is a convex set;
     if preferences are strictly convex and $p \gg 0$, $h(p, u)$ is a singleton
  3. "no excess utility" – if $u \geq u(0)$ and $p \gg 0$, then for any $x \in h(p, u)$, $u(x) = u$
- Proofs.
  1. $h(\lambda p, u)$ solves
     $$\min_{u(x) \geq u} (\lambda p) \cdot x \quad = \quad \lambda \min_{u(x) \geq u} p \cdot x$$
     the same minimization problem as $h(p, u)$, so it has the same solution
  2. if x and x' are both in $h(p, u)$, then $u(x) \geq u$, $u(x') \geq u$, and $p \cdot x = p \cdot x' = e(p, u)$
     if preferences are convex, $u(x^t) \geq u$ and $p \cdot x^t = p \cdot x$, so $x^t \in h(p, u)$ as well
     if preferences are strictly convex, then if $x' \neq x$, $u(x^t) > u(x)$, which would contradict "no excess utility" proven below
  3. If $u(x) > u$, then since u is continuous, there's some $t < 1$ with $u(tx) \geq u$
     - If this weren't the case, then consider the sequence $\{u(\frac{1}{2}x), u(\frac{3}{4}x), u(\frac{7}{8}x), \ldots\}$
     - Since $\{\frac{1}{2}x, \frac{3}{4}x, \ldots\} \to x$, if u is continuous, the sequence converges to $u(x)$, which by assumption was strictly greater than u
     - But then the later elements of this sequence must be close to $u(x) > u$, and therefore also above u

     But if for some $t < 1$, $u(tx) \geq u$, then the bundle tx, which is strictly cheaper than x, gives a cheaper way to achieve the target utility, so x couldn't solve the EMP
     (Also note this result doesn't require preferences to be LNS, just that u is continuous)

- **Proposition.** Suppose u is a continuous utility function representing **locally non-satiated** preferences on $\mathbb{R}^k_+$. Then $e(p,u)$ is...

  1. homogeneous of degree 1 in p: $e(\lambda p, u) = \lambda e(p,u)$
  2. continuous in p and u
  3. nondecreasing in p, strictly increasing in u if $p \gg 0$
  4. concave in p

- Proofs.

  1. Well,
  $$e(\lambda p, u) \quad = \quad \min_{u(x)\geq u} (\lambda p)\cdot x \quad = \quad \lambda \min_{u(x)\geq u} p\cdot x \quad = \quad \lambda e(p,u)$$
  or, we already showed $h(\lambda p, u) = h(p,u)$, and
  $$e(\lambda p, u) \quad = \quad (\lambda p)\cdot h(\lambda p, u) \quad = \quad (\lambda p)\cdot h(p,u) \quad = \quad \lambda e(p,u)$$
  2. We'll skip the proof, but the intuition is similar to continuity of $v(p,w)$ – if p changes a little, the min value changes a little; if u changes a little, the set $\{x : u(x) \geq u\}$ only changes a little since $u(\cdot)$ is continuous
  (This part does depend on preferences being LNS, though.)
  3. Suppose $p' > p$, and pick $x \in h(p', u)$. So $u(x) \geq u$, and
  $$e(p', u) \quad = \quad p'\cdot x \quad \geq \quad p\cdot x$$
  so $e(p,u) \leq p\cdot x \leq e(p',u)$.
  For u, let $u' > u$, and pick $x \in h(p,u')$, so $u(x) = u'$. If $e(p,u) \geq e(p,u') = p\cdot x$, then x would be a minimizer at (p,u), so $x \in h(p,u)$, violating no-excess-utility since $u(x) > u$.
  4. For concavity, we need to show
  $$e(tp + (1-t)p', u) \quad \geq \quad te(p,u) + (1-t)e(p',u)$$
  Pick $x \in h(tp+(1-t)p', u)$, so that $u(x) \geq u$.
  Then $e(p,u) \leq p\cdot x$, and $e(p',u) \leq p'\cdot x$
  So
  $$e(tp+(1-t)p', u) \quad = \quad (tp+(1-t)p')\cdot x \quad = \quad t(p\cdot x)+(1-t)(p'\cdot x) \quad \geq \quad te(p,u)+(1-t)e(p',u)$$
  and we're done.
  (More intuition: e is the min of a bunch of linear functions...)